

XP Architect

Formative Usability Test

Task 11 · Riga Technical University · 25/26-P

COURSE	Datorsistēmu projektēšanas pamati / Fundamentals of Computer Systems Design
INSTRUCTOR	Ksenija Lāce
PROJECT	XP Architect — player-experience design meta-game
TEAM LEADER	Temurbek Khaydarov · 231ADB143
DELIVERABLE	3 test scenarios · functional click-through prototype · 5-user test analysis · top-3 insights
SUBMITTED	Lecture 11 — Usability Testing

SCENARIOS

3

PARTICIPANTS

5

WIREFRAMES

12

TOP INSIGHTS

3

LIVE PROTOTYPE

<https://temur007222.github.io/xp-architect/>

Hosted on GitHub Pages · open in any modern browser (desktop or mobile) · responsive 360 – 1440 px.

1 · Team & Methodology

Team composition

The XP Architect team was registered in Task 6 (graded 10/10). The full team roster is reproduced below for traceability.

#	NAME	ROLE	STUDENT ID
1	Temurbek Khaydarov	Team Leader · usability lead · prototype	231ADB143
2	[Member 2 — replace]	Game design · scenario writing	[ID]
3	[Member 3 — replace]	Wireframing · prototype assistance	[ID]
4	[Member 4 — replace]	Test facilitator · user recruitment	[ID]
5	[Member 5 — replace]	Note-taker · session timing	[ID]
6	[Member 6 — replace]	Note-taker · post-session interviews	[ID]
7	[Member 7 — replace]	Data analysis · prioritisation matrix	[ID]
8	[Member 8 — replace]	Documentation · final report	[ID]

Note: placeholder rows are filled with the team members listed in Team members.pdf submitted for Task 6. Replace [Member X — replace] entries with the actual names before final submission.

Test methodology

We ran a **formative** moderated usability test — the goal was to surface design problems early, not to measure final performance. The protocol follows the lecture-11 outline:

Format	Moderated, in-person · think-aloud protocol
Participants	5 users (Nielsen rule — 5 users surface ~85% of usability issues)
Recruitment	Convenience sample · classmates and friends · all 18–28 yrs · mobile-app literate
Device	Functional click-through prototype hosted on GitHub Pages, viewed on a 390 × 780 phone-sized viewport (laptop)
Roles	1 facilitator (reads scenario, prompts) · 2 note-takers (timing + observations)
Per session	≈15 min: 2 min briefing · 3 × scenario · 3 min post-test interview
Measured	Completion rate · time-on-task · errors · qualitative comments
Success rule	Task is 'complete' if the user reaches the success state within 3 attempts and no facilitator help

Participant profiles

ID	AGE	PROFILE	MOBILE GAMING
P1	22	Casual gamer · CS student	Daily
P2	19	Hardcore PC gamer · rarely on mobile	Weekly
P3	27	Non-gamer · UX designer	Rarely
P4	24	Indie game developer · Unity	Daily
P5	21	RTU classmate · plays mobile puzzles	Daily

2 • Test Scenarios

Scenarios follow the format used in the lecture slides — each scenario sets a **context**, states a **goal**, lists a concrete **task**, defines **success criteria**, and notes the **screens involved**. They cover the three high-priority flows of XP Architect: onboarding, core gameplay loop, and post-game / progression.

SCENARIO 1 First-time onboarding	
Context	A friend recommended XP Architect to a player who has never used it before. They open the app for the first time and want to understand the concept before committing to a real session.
Goal	Discover what the game is about, learn how to play, and reach the first interactive gameplay screen.
Task given to user	<i>"You just downloaded XP Architect because a friend told you it's interesting. Sign in, figure out what the game is about, and start your first game session. Tell me out loud what you're thinking as you go."</i>
Success criteria	User reaches the Place Events screen (W4) with a chosen archetype, having visited either the Home screen (W2) or the How-to-Play tutorial (W12).
Screens involved	W1 Login · W2 Home/About · W12 Tutorial · W3 Timeline Setup · W4 Place Events
Expected flow	W1 → W2 → (W12 optional) → W3 → W4

SCENARIO 2 Design a complete game session	
Context	A returning player has 5 minutes between classes and wants to design one full satisfaction curve for a Casual archetype player, then check the final score.
Goal	Complete the core gameplay loop — pick archetype, place 8 events on the timeline, finalise, and review the session summary.
Task given to user	<i>"You're back in the app. Design a complete game session for a Casual player archetype. Place all 8 events, lock in the last one, and tell me your final satisfaction score."</i>
Success criteria	User reaches the Summary screen (W6) showing a satisfaction score, having placed at least one event on the timeline and confirmed a final round event.
Screens involved	W2 Home · W3 Timeline Setup · W4 Place Events · W5 Final Round · W6 Summary
Expected flow	W2 → W3 → W4 → W5 → W6

SCENARIO 3 Check progress and adjust settings	
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Context	A player has just finished a session, is curious how they rank against other players, wants to look at their session history, and decides to lower the difficulty for the next round.
Goal	Navigate post-game — find the leaderboard, browse past sessions, and change a gameplay setting.
Task given to user	<i>"You just finished a game. Find out how you rank on the leaderboard, look at your past sessions, then change the difficulty to make the next round shorter."</i>
Success criteria	User opens W11 Settings and interacts with the Difficulty selector, having visited at least one of W8 Leaderboard or W10 History first.
Screens involved	W6 Summary · W7 Menu · W10 History · W8 Leaderboard · W9 Profile · W11 Settings
Expected flow	W6 → W7 → W10 → W8 → W11

3 · Functional Click-Through Prototype

SUBMISSION LINK

<https://temur007222.github.io/xp-architect/>

Source repository: <https://github.com/temur007222/xp-architect> (public).

What changed since the wireframe version

The original Task-10 wireframe prototype was a single static HTML click-through. After running the formative test, we re-built it as a **functional React + TypeScript app** (Vite + Tailwind + @dnd-kit + Recharts + Zustand) so that scenarios behave like the real product:

- **Real drag-and-drop** on W4 using @dnd-kit (PointerSensor for mouse, TouchSensor for mobile) — every tile placement triggers a live recompute of the satisfaction curve.
- **Real scoring** per §3.4 of the spec — RMSE between player and target curve → 0–100 score → Bronze / Silver / Gold rank tier. Covered by a vitest unit suite (11 tests).
- **Persistent sessions** via localStorage (key xp-architect-v1) — profile, sessions, badges, and settings survive reloads.
- **All three top-priority usability fixes** from this test are implemented in the live prototype (see §5).

Hosting

OPTION	HOW	PUBLIC LINK?
GitHub Pages USED	Source on main, built with <code>npm run build</code> , deployed via <code>gh-pages -d dist</code> . Free static hosting.	Yes
Open locally	Clone repo · <code>npm install</code> · <code>npm run dev</code> · open <code>http://localhost:5173</code> . Best for offline review.	No
Netlify Drop	Drag-and-drop the <code>dist/</code> folder at <code>app.netlify.com/drop</code> . Gets a public URL in <30 seconds.	Alt.
Vercel	Connect the GitHub repo at <code>vercel.com/new</code> — auto-detects Vite, deploys on every push.	Alt.

Screens included

W1 Login / Welcome	W5 Final Round placement	W9 Player Profile
W2 Home / About game	W6 Game Summary + insights	W10 Player History
W3 Timeline Setup (archetype picker)	W7 Navigation Menu (overlay drawer)	W11 Game Settings
W4 Place Events (core loop, with live satisfaction graph & drag-drop)		W12 How to Play (4-step tutorial)
W8 Leaderboard	W10b History detail (per-session)	<i>13 routable screens total.</i>

Changes from Task 10 wireframes

- **Functional** archetype selection on W3 — three tap-targets drive the target satisfaction curve for the rest of the run.
- **Live** satisfaction graph on W4 with prominent target overlay (B1 fix) — curve recomputes within ~10ms of each placement.
- **Confirmation flow** on W5 — primary teal Lock button + irreversibility modal (B3 fix); warning colour moved to a banner.
- **Insight cards** on W6 — Strongest, Weakest, and Closing moments auto-generated from the curve delta.
- **Menu drawer** (W7) — global overlay reachable from every top bar.
- **Tappable settings rows** (W11) — chevron + bordered button style for every pill selector (C2 fix).
- **History detail view** (W10b) — tapping a row opens that session's timeline + curve, not a generic Summary loop (C3 fix).

4 • Test Results

Each scenario is reported with the three measures requested: **completion rate**, **time to complete** (mean of successful sessions), and **three key insights** distilled from observation notes and post-test interviews. Times below are mm:ss.

SCENARIO 1 • RESULTS			First-time onboarding
COMPLETION RATE		MEAN TIME	PARTICIPANTS
5 / 5 (100%)		01:16	5

Per-participant data

ID	COMPLETED?	TIME	NOTES (THINK-ALOUD EXCERPTS)
P1	✓ Yes	00:47	"Sign in, then I guess this big card is Start. OK." Skipped tutorial.
P2	✓ Yes	01:12	"What's an archetype?" — paused 4s on W3, picked Casual by default.
P3	✓ Yes	01:38	Read About section thoroughly. Tapped How to Play before starting — only user who did.
P4	✓ Yes	00:39	Fastest. Skipped everything: "I just want to see the game."
P5	✓ Yes	02:04	Hesitated on W1 — "do I sign in or create account, are these different?" Tapped Create, then went back.

Three key insights

A1	Sign In vs Create Account has no visual hierarchy P3 and P5 hesitated on W1, unsure which button matched their situation. Both buttons have identical weight. Suggest: make Sign In primary, Create secondary text-link below.	Severity 2	Frequency 2
A2	How to Play is skipped by 4 of 5 users The 'Start Designing' card is bigger, more colorful, and above the How-to-Play button. Only P3 opened the tutorial. Users then struggled on W3 not knowing what an 'archetype' meant. Suggest: surface a 30-second onboarding overlay on first-ever launch.	Severity 2	Frequency 3
A3	Pre-filled login fields confuse first-time users P2 and P5 saw 'player@rtu.lv' already typed in and asked: 'Am I already logged in?' The prototype's placeholder text reads as real data. Cosmetic but affects trust on first impression.	Severity 1	Frequency 2

SCENARIO 2 • RESULTS			Design a complete game session
COMPLETION RATE		MEAN TIME	PARTICIPANTS
4 / 5 (80%)		04:52	5

Per-participant data

ID	COMPLETED?	TIME	NOTES (THINK-ALOUD EXCERPTS)
P1	✓ Yes	04:12	"Cool, I see numbers go up." Did not notice the dashed target line until prompted.
P2	✓ Yes	03:45	Mastered drag pattern fast. Confused by orange Lock button: "Wait, is this an error?"
P3	✓ Yes	06:22	Slowest. Methodically tried each event type. Asked twice: "What's a good score? What am I aiming for?"
P4	✓ Yes	05:08	Ignored the satisfaction graph entirely for 4 of 8 placements. Realised at round 6 there was a curve.
P5	✗ No	—	Gave up at round 5/8: "I don't understand if I'm doing well or badly. This is just numbers."

Three key insights

B1	Dashed target curve is invisible — users don't know what they're aiming for All 5 users at some point asked 'what am I aiming for?'. The pink dashed target overlay was too thin and visually similar to the score line. P5 abandoned the task because of this. This is the single highest-impact issue.	Severity 3	Frequency 3
B2	'Round 3/8' header not perceived as progress indicator P3 and P4 asked 'how many more rounds are there?' even though the answer was at the top of the screen. The label is small (14px), placed in the top bar where users don't look. Suggest: add a visible progress bar.	Severity 2	Frequency 2
B3	Orange 'Lock final event' button reads as error / warning P1, P2, and P4 all paused before tapping. P2 explicitly said it looked like an error message. The warning colour was meant to signal irreversibility but reads as 'do not press'. Suggest: keep button teal, move the irreversibility warning into a confirmation modal.	Severity 2	Frequency 3

SCENARIO 3 · RESULTSCheck progress and adjust settings

COMPLETION RATE
4 / 5 (80%)

MEAN TIME
02:38

PARTICIPANTS
5

Per-participant data

ID	COMPLETED?	TIME	NOTES (THINK-ALOUD EXCERPTS)
P1	✓ Yes	01:54	Found menu fast. Tapped Difficulty pill on first try.
P2	✓ Yes	02:12	Looked for Settings gear icon first; eventually used the hamburger menu.
P3	✓ Yes	03:08	Tapped Profile avatar first looking for settings; navigated to W11 via the gear in W9 header.

ID	COMPLETED?	TIME	NOTES (THINK-ALoud EXCERPTS)
P4	✓ Yes	01:38	Fastest. Knew to look in menu drawer.
P5	✗ No	—	Could not find Settings — tried Profile, History, Leaderboard, gave up. "Where would settings even be?"

Three key insights

C1	Hamburger menu (≡) icon discoverability is weak P2 and P3 looked for a gear icon or bottom-tab bar before noticing the hamburger. P5 never found it. Mobile users increasingly expect bottom-tab navigation. Suggest: add a 4-item bottom bar (Home · Play · Stats · Profile) and keep the menu as secondary.	Severity 2	Frequency 2
C2	Settings pills look decorative — users don't realise they're tappable 'Normal · 8' and 'Configure' pills (W11) have no affordance suggesting they open a picker. P1 tapped successfully but P3 and P5 read them as labels. Suggest: add a chevron (›) and use a bordered button style.	Severity 2	Frequency 3
C3	Tapping a History row navigates back to the Summary — feels like a dead-end loop P3 and P4 tapped multiple history rows expecting a detailed timeline view, but each tap led back to the same generic Summary (W6). The history items should open a read-only past-session detail view.	Severity 1	Frequency 3

5 · Top 3 Insights — Prioritisation

Method: Severity × Frequency matrix

We use the prioritisation method covered in the lecture: each observed issue is scored on two axes, then ranked by their product. This is the standard formative-test triage used in Nielsen Norman Group research and matches the lecture-11 framework.

Severity	1 — Cosmetic (annoyance, no functional impact) 2 — Major (slows task, requires workaround) 3 — Critical (blocks task, user gives up or fails)
Frequency	1 — Rare (1 user affected) 2 — Common (2–3 users affected) 3 — Universal (4–5 users affected)
Priority	Priority = Severity × Frequency → range 1 to 9. Tie-breaker: higher severity wins; second tie-breaker: affects the core gameplay loop.

All 9 insights scored

RANK	CODE	INSIGHT (SHORT)	SCENARIO	SEV	FREQ	PRIORITY
1	B1	Target curve invisible	S2	3	3	9
2	A2	Tutorial skipped by default	S1	2	3	6
3	B3	Final-event button warning-colour confusion	S2	2	3	6
4	C2	Settings pills not visually tappable	S3	2	3	6
5	A1	Sign-in / Create has no hierarchy	S1	2	2	4
6	B2	Round counter not seen as progress	S2	2	2	4
7	C1	Hamburger menu weak discoverability	S3	2	2	4
8	C3	History row → Summary loop	S3	1	3	3
9	A3	Pre-filled login fields confuse	S1	1	2	2

Three insights tie at priority = 6 (A2, B3, C2). Applying the tie-breaker — severity then impact on the core loop — A2 and B3 rank above C2 because they affect every new user (A2) or the core gameplay action (B3); C2 affects an optional configuration path. All three are listed in the final top 3 because they share the same priority score. **All four issues at priority ≥ 6 are implemented in the live prototype.**

Selected top 3 — implemented in the live prototype

#1 · B1 · Priority 9	Scenario 2 · Severity Critical · 3 · Frequency Universal · 3
Target satisfaction curve is invisible	

What we saw: The dashed target overlay on the satisfaction graph (W4) is too thin and visually similar to the user's own score line. All 5 users at some point asked 'what am I aiming for?' — P5 abandoned the entire scenario because of this. The game's core feedback loop fails when users can't see the goal.

Fix shipped: Target curve is now **3 px pink-acc dashed**, with a labelled TARGET callout at the rightmost data point and a one-line caption above the graph: *'Match the dashed line for the best score.'* See `src/components/SatisfactionChart.tsx`.

#2 · A2 · Priority 6

Scenario 1 · Severity Major · 2 · Frequency Universal · 3

How to Play is skipped by 4 of 5 first-time users

What we saw: On W2, the 'Start Designing' CTA is larger, brighter, and placed above the 'How to Play' button. Only P3 (a UX designer) opened the tutorial. The others then hit W3 not knowing what an 'archetype' was, slowing their session and creating uncertainty that compounded in Scenario 2.

Fix shipped: On the very first launch only, the app surfaces a **3-step onboarding modal** before W2 becomes interactive. The flag `player.hasSeenOnboarding` is persisted to `localStorage` so it never appears again. See `src/components/OnboardingModal.tsx`.

#3 · B3 · Priority 6

Scenario 2 · Severity Major · 2 · Frequency Universal · 3

Orange 'Lock final event' button reads as an error

What we saw: On W5, the warning-orange Lock button was meant to signal irreversibility. Instead, 3 of 5 users hesitated, and P2 explicitly said 'wait, is this an error?'. The warning colour creates hesitation at the most important confirm action of the entire game.

Fix shipped: The Lock button is now **primary teal** (confirm semantics). The irreversibility warning lives in an orange banner above the chart, and the actual confirmation is a `ConfirmModal` shown after tap (*'Lock the satisfaction curve? ... Yes, lock it'*). See `src/screens/W5Final.tsx`.

6 • Reflection

What the test confirmed

The core mechanic of XP Architect — placing events on a timeline and watching a curve respond — is fundamentally engaging. All 5 users understood the drag-place metaphor within 30 seconds of reaching W4. The information architecture and navigation system designed in Task 8 also held up: the menu drawer (W7) covered every screen the users needed to reach.

Where the design failed

Two patterns emerged. First, **visual hierarchy of feedback**: users don't see what they should be optimising for (insight B1). The satisfaction graph is the most important UI element of the game and its target overlay needs to be impossible to miss. Second, **colour semantics in CTAs**: using warning-orange for the most important confirm button (B3) sends the wrong signal — colour must match action, not consequence.

Methodology limitations

- **Convenience sample.** All participants were 18–28 and mobile-app literate. Older users or first-time smartphone users may surface different issues, especially around the hamburger menu (C1).
- **Single session per user.** Returning-user behaviour (e.g. menu discoverability after 10 sessions) is speculative based on first-encounter data.
- **Self-reported times.** Stopwatch start/stop was facilitator-controlled and may have ± 2 seconds of error per measurement. The qualitative observations are more reliable than the time figures alone.
- **Previously: click-through-only prototype.** The original wireframe version faked the drag-and-drop. For the next round of testing we now have a **fully functional prototype** (live at the URL above) so the timeline interaction will be evaluated on real touch input.

What was shipped after this test

-  All three top-priority fixes (**B1**, **A2**, **B3**) implemented in the live React build.
-  Three additional priority-4 / priority-6 fixes folded in: **A1** (login hierarchy), **B2** (progress bar on W4), **C2** (chevron + bordered settings rows), **C3** (per-session history detail view).
-  Deferred for v2: **C1** bottom-tab navigation (low cost, but would change information architecture — to be validated in next test).
-  Deferred for v2: **A3** placeholder text on login (cosmetic).

Next steps

- Re-test the same scenarios with 5 fresh participants on the live prototype to confirm the fixes work and don't introduce new issues — particularly around the new onboarding overlay (A2) and the confirm modal (B3).
- Validate the W4 drag-and-drop on real mobile devices (the earlier test was on a laptop emulating a phone viewport — now that drag is functional we can run real touch testing).
- Decide on the bottom-tab navigation question (C1) in the next iteration based on observed behaviour.

